

Ay3 Inferencia

P4) Ayudante pasado.

$$x_1, \dots, x_n \sim f(x|\alpha, \beta) = \frac{2\beta^2 x}{\alpha^2} \mathbb{1}_{(0, \alpha/\beta)}(x)$$

$\alpha, \beta > 0$

a) $f(\vec{x}|\alpha, \beta) = 2^n \left(\frac{\beta}{\alpha}\right)^{2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \alpha/\beta)}(x_i)$ $\Theta = \{\alpha, \beta : \alpha > 0, \beta > 0\}$

b) Identificabilidad

$$\vec{x} = \{x \in \mathbb{R}^n : x_i > 0, x < \frac{\alpha}{\beta}\}$$

$\frac{\beta}{\alpha} = k \rightarrow \beta = 2, \alpha = 2 \Rightarrow \beta = 1, \alpha = 1$ no es identificable

c) Sea $\Theta = \alpha/\beta$

$$f(\vec{x}|\Theta) = 2^n \Theta^{-2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \Theta)}(x_i)$$

$$f(\vec{x}|\Theta_1) = f(\vec{x}|\Theta_2)$$

$$2^n \Theta_1^{-2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \Theta_1)}(x_i) = 2^n \Theta_2^{-2n} \prod_{i=1}^n x_i \mathbb{1}_{(0, \Theta_2)}(x_i)$$

$$\left(\frac{\Theta_1}{\Theta_2}\right)^{-2n} \frac{\prod_{i=1}^n x_i \mathbb{1}_{(0, \Theta_1)}(x_i)}{\prod_{i=1}^n x_i \mathbb{1}_{(0, \Theta_2)}(x_i)} = 1$$

$$\Rightarrow \Theta_1 = \Theta_2$$

Teorema (Suficiencia)

$f(\vec{x}|\theta, T(\vec{x}))$ es independiente de θ vs T

P5) $x_1, \dots, x_n \sim \text{Poisson}(\theta)$ $T(\vec{x}) = \sum_{i=1}^n x_i$

$$z = x_1 + x_2$$

$$P_r(z=z) = P_r(x_1 + x_2 = z)$$

$$= \sum_{x_1=0}^z P_r(x_1 = z - x_2 | x_2 = x_2) P_r(x_2 = x_2)$$

$$= \sum_{x_2=0}^z \frac{e^{-\theta} \theta^{z-x_2}}{(z-x_2)!} \cdot \frac{e^{-\theta} \theta^{x_2}}{x_2!}$$

$$= \sum_{x_2=0}^z \frac{e^{-2\theta} \theta^z}{(z-x_2)! x_2!} = \sum_{x_2=0}^z \frac{z!}{z! (z-x_2)! x_2!} \theta^z = \frac{z!}{z!} \theta^z = \theta^z$$

$$x_i \sim f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

CP2)

$$E(e^{xt}) = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} \cdot e^{xt} = \sum_{x=0}^{\infty} \frac{e^{-\lambda} (\lambda e^t)^x}{x!} = e^{\lambda(e^t - 1)}$$

$$= e^{\lambda(e^t - 1)} = e^{\lambda(e^t - 1)}$$

Poisson(λe^t)

for Indicator

$$T(\vec{X}) \sim \text{Poisson}(n\theta) \quad \frac{PD}{\text{, no depend de } \theta}$$

$$f(\vec{X}=x | T(\vec{X})=T) = \frac{f(\vec{X}|\theta)}{f(T|\theta)}$$

$$= \frac{\prod_{i=1}^n e^{-\theta} \theta^{x_i}}{x!}$$

$$= \frac{e^{-n\theta} \theta^{\sum x_i}}{\prod_{i=1}^n x_i!}$$

$$= \frac{\theta^{\sum x_i}}{\prod_{i=1}^n x_i!} = \frac{\theta^{\sum x_i}}{n^{\sum x_i} \theta^{\sum x_i}} = \frac{1}{n^{\sum x_i}}$$

$$= \frac{T!}{\prod_{i=1}^n x_i! \cdot n^{\sum x_i}}$$

no depend de

OP2

$$Z = X_1 + X_2$$

$$E(e^{z\tau}) = E(e^{x_1\tau}) \cdot E(e^{x_2\tau})$$

$$= e^{\lambda(e^\tau-1)} \cdot e^{\lambda(e^\tau-1)}$$

$$= e^{2\lambda(e^\tau-1)}$$

$$\rightarrow Z \sim \text{Poisson}(2\lambda)$$

P2 $x_1, \dots, x_n$ $\sim$ independent $f_i(x_i|\theta)$

Theorem Factorization: $f(\vec{x}|\theta) = g(T(x)|\theta) h(x)$

$g \geq 0 \Rightarrow T$ est suffisant

$$f_i(x|\theta) = \begin{cases} \frac{1}{2i\theta} & -i(\theta-1) < x < i(\theta+1) \\ 0 & \text{c.o.c} \end{cases}$$

$$f(\vec{x}) = \prod_{i=1}^n \frac{1}{2i\theta} \mathbb{1}_{[-i(\theta-1), i(\theta+1)]}(x_i)$$

$$2^{-n} \theta^{-n} \prod_{i=1}^n \frac{1}{i} \mathbb{1}_{(-i(\theta-1), i(\theta+1))}(x_i)$$

$$2^{-n} \theta^{-n} \left(\prod_{i=1}^n \frac{1}{i} \right) \mathbb{1}_{\left(\min\left(\frac{x_i}{i}\right) > -(\theta-1) \right)} \mathbb{1}_{\left(\max\left(\frac{x_i}{i}\right) < \theta+1 \right)} \mathbb{1}(\vec{x})$$

$$2^{-n} \frac{1}{n!} \theta^{-n} \mathbb{1}_{\left(\min\left(\frac{x_i}{i}\right) > -(\theta-1) \right)}$$

$$h(\vec{x})$$

$$g(T(\vec{x})|\theta)$$

$$\begin{aligned} & -i(\theta-1) < x_i \\ & \min(x_i) \\ & -(\theta-1) < \frac{x_i}{i} \\ & -(\theta-1) < \min(x_i/i) \\ & x < i(\theta+1) \\ & \frac{x}{i} < \theta+1 \\ & \max\left(\frac{x_i}{i}\right) < \theta+1 \end{aligned}$$

$$T = \left[\min\left(\frac{x_i}{i}\right), \max\left(\frac{x_i}{i}\right) \right]$$

$g \geq 0 \Rightarrow T$ est suffisant

Def: Superficie minimal

i) T es superficie por σ

ii) T es función de cualquier otra estadística suficiente

En palabras simples, T es minimal si no se puede seguir comprimiendo la información.

Teorema Sea $x, y \in X$ espacio muestral

$\frac{f(x|\theta)}{f(y|\theta)} = k$ no depende de $\theta \Leftrightarrow T(x) = T(y)$, entonces, $T(X)$ es una estadística suficiente minimal.

P) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, \theta)$ $\alpha, \theta > 0$

$$f(x_i|\alpha, \theta) = \frac{1}{\Gamma(\alpha) \theta^\alpha} x_i^{\alpha-1} e^{-x_i/\theta}$$

$$f(\vec{X}|\alpha, \theta) = [\Gamma(\alpha)]^{-n} \theta^{n\alpha} \left[\prod_{i=1}^n x_i \right]^{\alpha-1} e^{-\sum x_i/\theta}$$

$$\frac{f(\vec{X}|\alpha, \theta)}{f(\vec{Y}|\alpha, \theta)} = \frac{[\Gamma(\alpha)]^{-n} \theta^{n\alpha} \left[\prod_{i=1}^n x_i \right]^{\alpha-1} e^{-\sum x_i/\theta}}{[\Gamma(\alpha)]^{-n} \theta^{n\alpha} \left[\prod_{i=1}^n y_i \right]^{\alpha-1} e^{-\sum y_i/\theta}}$$

$T(X) = \left(\prod x_i, \sum x_i \right)$

$$= \left[\frac{\prod_{i=1}^n x_i}{\prod_{i=1}^n y_i} \right]^{\alpha-1} e^{-\frac{1}{\theta} (\sum x_i - \sum y_i)}$$

no depende de α, θ

solo si:

$$\frac{\prod x_i}{\prod y_i} = 1 \Rightarrow \prod_{i=1}^n x_i = \prod_{i=1}^n y_i \quad \wedge \quad \sum x_i = \sum y_i$$

$$T_1(X) = T_1(Y) \quad \wedge \quad T_2(X) = T_2(Y)$$

P41

$$X_i \sim \text{Geo}(\theta_i)$$

$$\theta_1 = \theta$$

$$\theta_2 = 2\theta$$

$$X \sim \text{Geo}(\theta)$$

$$P(X=x) = \theta(1-\theta)^{x-1}$$

$$n=1 \quad P(\vec{X} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}) = \theta_1 (1-\theta_1)^{x_1-1} \theta_2 (1-\theta_2)^{x_2-1}$$

$$= \theta_1 \theta_2 (1-\theta_1)^{x_1-1} (1-\theta_2)^{x_2-1}$$

$$= 2\theta^2 (1-\theta)^{x_1-1} (1-2\theta)^{x_2-1}$$

$$R_r(\vec{X} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}) = \prod_{i=1}^n 2\theta^2 (1-\theta)^{x_{1i}-1} (1-2\theta)^{x_{2i}-1}$$

$$= \underbrace{2^n}_{h(x)} \underbrace{\theta^{2n} (1-\theta)^{\sum x_{1i} - n} (1-2\theta)^{\sum x_{2i} - n}}_{g(T(x) | \theta)}$$

$$T(x) = (\sum x_{1i}, \sum x_{2i})$$

$$\frac{f(\vec{x})}{h(\vec{x})} = \frac{2^n \theta^{2n} (1-\theta)^{\sum x_{1i} - n} (1-2\theta)^{\sum x_{2i} - n}}{2^n \theta^{2n} (1-\theta)^{\sum y_{1i} - n} (1-2\theta)^{\sum y_{2i} - n}}$$

$$\sum x_{1i} = \sum y_{1i} \quad \sum x_{2i} = \sum y_{2i}$$